

Real-Time Fruit Detection and Counting

Hassina Ait Issad

LARI Laboratory

Mouloud Mammeri University

Tizi-Ouzou, Algeria

hassina.aitissad@ummto.dz

Massyl Lakef

dept. Computer science

Mouloud Mammeri University

Tizi-Ouzou, Algeria

lakmassyl1@gmail.com

Salim Boucherk

dept. Computer science

Mouloud Mammeri University

Tizi-Ouzou, Algeria

salim.boucherk@fgei.ummto.dz

Karima Chemoun

LARI Laboratory

Mouloud Mammeri University

Tizi-Ouzou, Algeria

karima.chemoun@ummto.dz

Sarah Oubabas

LARI Laboratory

Mouloud Mammeri University

Tizi-Ouzou, Algeria

sarah.oubabas@ummto.dz

Rachida Aoudjit

LARI Laboratory

Mouloud Mammeri University

Tizi-Ouzou, Algeria

rachida.aoudjit@ummto.dz

Abstract—In agriculture, detecting and counting fruits are prominent tasks that facilitate harvesting and yield estimation. Since traditional methods such as manual counting are tedious and error-prone, developing a real time system to automatically and accurately detect, classify, and count fruits in an agricultural environment is required. The aim of this paper is to propose a system based on cutting-edge technologies, YOLOv8 and ByteTrack, to accurately detect and count fruits in real time. Consequentially, farmers will have access to precise information about the quantity and variety of fruits in their fields, enabling them to make informed decisions and maximize the profitability of their agricultural operations. Furthermore, the system will enable them to enhance the efficiency of agricultural work and reduce the required workforce. In this paper, a dataset containing 20 classes of fruits with diverse backgrounds and lighting conditions is constructed to train and assess the model's performance. The validation results of the detection and counting model showed an mAP 0.5 value of 94.2%, a recall of 87.6%, and a precision of 91.5%. The comparison between the manual counting and the model showed an accuracy of 75.33%.

Index Terms—Object detection, deep learning, multi-object tracking, smart agriculture, object counting.

I. INTRODUCTION

Agriculture represents one of the most crucial sectors of the economy, providing food and materials for numerous industries. This sector is constantly evolving and actively seeking innovative means to enhance production efficiency in order to meet the dietary demands of a growing population, estimated to surpass 9.6 billion by the year 2050 [1]. Consequently, the implementation of intelligent agricultural practices has become an indispensable necessity to achieve these food security objectives [2].

Fruit recognition and classification, as well as accurate counting, play an essential role in farm management. Accurate fruit recognition and precise counting of fruit will enable yields to be assessed, i.e., to obtain valuable information on the expected yields of different types of fruits. This will enable farmers to make informed decisions about maximizing

farm profitability. However, traditional methods are often time-consuming and prone to human error. This is why the integration of cutting-edge technologies such as Deep Learning (DL) opens up new prospects for automating and improving these tasks. Fruit counting can be classified into two main types: regression-based fruit counting and detection-based fruit counting. According to [3], the latter is considered to be more effective. In the case of fruit counting based on detection techniques, the accuracy of fruit detection and localization plays an important role in determining model performance.

Detecting fruit in agricultural settings has presented a significant challenge due to the intricate nature of growing environments. Initially, attempts have focused on extracting characteristics such as color, shape, or dominant features, followed by techniques like wavelet transform-based analysis and manual observation factors [2]- [4]. However, these methods are still inadequate for dealing with difficult situations like complex backgrounds, changes in lighting, and weather variations.

DL has shown significant progress in different fields, leading to the development of object detection models. These models are categorized into one-stage and two-stage models. The latter, such as Faster-RCNN [5], R-CNN [6], Fast-RCNN [7], and Mask R-CNN [8], show better accuracy and recall, while one-stage models, like SSD [9] and YOLO series [10], exhibit faster inference speed. YOLO is one of the latest advancements in DL algorithms used in object detection, which stands for You Only Look Once [10]. YOLO has proven to be highly efficient in detecting and recognizing objects across different contexts, making it an ideal choice for fruit detection applications. Various articles discuss the use of these algorithms for fruit detection, specifically focusing on tomatoes and apples in different conditions [11], [12]. New models like YOLO-Tomato [13], YOLO with DSConv, DPAG, and FEM [14], and TDPPL-Net [15] have been proposed to improve tomato detection precision in agriculture applications.

On the other hand, the progress of DL in the past few years has also greatly influenced the field of Multi-Object Tracking (MOT), which serves as a subsequent task to object detection. Multi-object tracking represents a crucial task in the field of computer vision. It focuses on the ability to locate and track objects in videos or image sequences over a period of time. This task holds immense significance for various real-world applications, such as robotics and autonomous driving. However, object tracking encounters several obstacles, such as dealing with changes in the appearance of objects, camera movement, occlusion, and alterations in lighting conditions. To overcome these challenges, advancements in DL have played a crucial role. These advancements have resulted in more robust and accurate tracking algorithms, enabling us to track objects with greater precision and reliability. Notable MOT algorithms include the Simple Online and Real-time Tracking (SORT) algorithm [16], the Deep-SORT algorithm, which is an extension of SORT [17], FairMOT [18], and ByteTrack [19]. ByteTrack, being one of the latest MOT algorithms, has demonstrated superior performance in soccer videos and satellite imagery compared to other advanced multi-object trackers [20], [21].

In the field of fruit counting, Egi et al. [22] focused on the use of YOLOv5 and Deep Sort algorithms, alongside a dataset of greenhouse tomatoes captured by drones. The dataset comprised three distinct categories, namely flowers, green tomatoes, and red tomatoes. This dataset was used to train and assess the counting model. At epoch 96, the optimal model for all classes is found with an accuracy of 0.618 at mAP 0.5. The average F1 score for the three classes was 0.63. The count results obtained manually and through the AI-Drone indicated that the accuracy rates for the three classes varied from 50% to 99%. Zhang et al. [23] presented a deep-learning-based algorithm designed for counting citrus fruits using video sequences. The algorithm comprised two sub-algorithms: OrangeYolo, used for detecting fruits, and OrangeSort, employed for tracking fruits. OrangeYolo achieved higher mean average precision values compared to other algorithms, while OrangeSort mitigated the issue of double-counting occluded fruits. Parico and Ahamed [24] developed a pear fruit counter using YOLOv4 models and the Deep SORT algorithm for mobile applications. YOLOv4-CSP was found to be the optimal model in terms of accuracy, while YOLOv4-tiny was ideal in terms of speed and computational cost. Hu et al. [25] introduced a new model that combined the self-attentive mechanism of vision transform with Yolov7 to overcome the difficulties encountered in accurately counting fruit yield within apple orchards. This model incorporated two attention mechanisms, CA and CBAM, and was trained and evaluated using apple images. The results showed an mAP of 91.3% and an F1 score of 0.85. Furthermore, the paper integrated two multi-objective tracking methods, namely SORT and Cascade-SORT, to allow for the counting of fruits in a variety of complicated situations across video frames. The results showed that the presented model had the potential to enhance fruit yield assessment techniques and decision-making processes in the fruit industry.

To the best of our knowledge, there are only a few studies in the literature that combine multi-object tracking algorithms with DL object detection algorithms for fruit detection and counting [22]- [25]. This area of research is still relatively unexplored and primarily focuses on a single type of fruit. The aim of this paper is to combine YOLOv8 and ByteTrack, which are currently recognized as the most advanced object detection models and MOT algorithms, respectively, to detect and count 20 different types of fruits. This research presents two key contributions. Firstly, a dataset consisting of fruit images under various conditions, including illumination variations and complex backgrounds, was collected and annotated. Secondly, the cutting-edge YOLOv8 object detection model and the ByteTrack multi-object tracking model were combined to accurately detect and count 20 different types of fruits.

II. METHODOLOGY

A. General Methodological Overview

Our proposed system involves six successive stages (Fig. 1). In the first stage, we collect fruit images, which are used to train and evaluate the YOLO model. Then, in the second stage, we subject the dataset to a treatment process involving annotation, augmentation and pre-processing, which increases the number of samples available in the dataset. The third step is to train the YOLOv8 object detection model on our dataset. Using the split dataset for validation purposes, we evaluate the refined model's detection performance and analyze the results obtained. Finally, in the last phase, we use the developed model to build a system capable of detecting and counting fruits in the field.

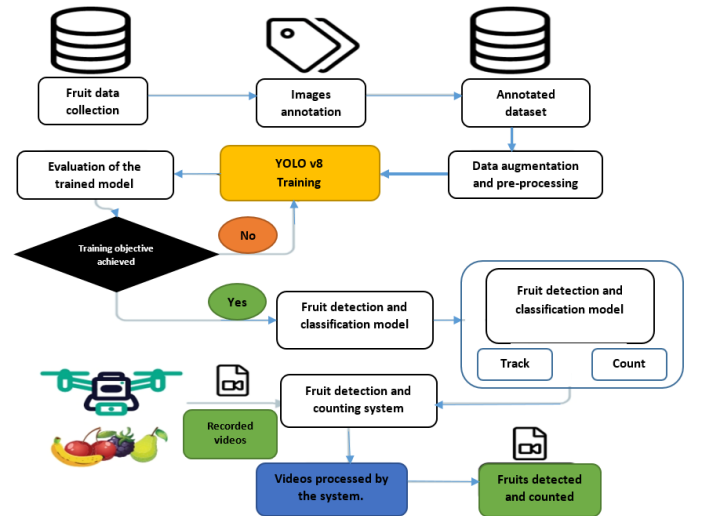


Fig. 1: Diagram of the proposed system.

B. Data collection

In order to train and validate the fruit detection and counting system illustrated in Fig. 1, we used a dataset representing various types of fruits, collected from gardens and fruit and vegetable markets. We also used the Internet as a source for

collecting images. This study focuses on the identification and detection of 20 fruit categories. During the data collection process, we began our research by consulting two databases, Kaggle and iStock, and the Google search engine. In all, we collected 746 images in a single JPG format, which we then divided into 20 distinct classes.

C. Data Annotation and preprocessing

Image annotation is an important step in the process of preparing the YOLO model for training. This annotation involves extracting features from an image and associating them with labeled inputs. During training, the machine can thus learn to recognize the features present in the labeled images. Consequently, the accuracy of the labeling has a major impact on the model accuracy. The YOLO model must be able to learn distinctive features for different types of fruits. In the annotation process, the type and location of fruits present in the labelled image are specified. To this end, Roboflow's Annotate tool was used to label the images. Information on the location and category of fruits in the image was retrieved and saved as a text file in YOLO format.

To improve model performance, Roboflow offers a data pre-processing process, which was applied to the dataset. For the sake of uniformity, the images were resized to 640 640 pixels. Automatic orientation was also used to ensure that image orientation has no undesirable impact on the model.

D. Data augmentation and splitting

In general, Deep Learning models tend to perform better when more data are available. Besides, to counteract over fitting and enable the model to generalize better, increasing the number of training samples can be an effective strategy. In this study, additional training data were generated by applying various image enhancement techniques, including flipping, horizontal and vertical rotation, brightness modification, grayscale conversion and blurring. Fig. 2 shows the application of data augmentation techniques to fruit images.

After the data augmentation stage, the dataset reached a total of 1792 samples, divided into 20 different fruit classes. To train the fruit detection model efficiently, the dataset was split into training, validation, and test sets. Specifically, the dataset was split into 1569 images for training, 153 images for validation, and 70 images for testing.

E. Description of the YOLO Model

YOLOv8 is the latest version of the YOLO family, launched by Ultralytics on 10 January 2023 [26]. It is an advanced and cutting-edge model that outperforms its predecessors in the YOLO series in terms of both detection accuracy and speed [26]- [28]. Unlike its predecessors, YOLOv8 implements an anchor free detection technique that predicts the center of an object, rather than predicting its distance from a known anchor box. This technique reduces the number of box predictions, thus accelerating the post-processing process.

The YOLOv8 architecture comprises three key components: the backbone, the neck, and the head. This architecture is



Fig. 2: Different data augmentation techniques applied to fruit images.

illustrated in Fig. 3. The backbone of YOLOv8 is in charge of extracting feature maps from input images. It is based on a modified CSPDarknet-53 that uses a series of convolutional layers enabling the network to learn complex representations [28]. The neck acts as a bridge between the backbone and the head, facilitating the integration of multi-scale features. YOLOv8 uses a Path Aggregation Network - Feature Pyramid Network (PAN-FPN) structure as its neck module, allowing the model to merge features from different levels of the backbone network [28]. The head of YOLOv8 processes the feature maps extracted to generate bounding box predictions and class probabilities.

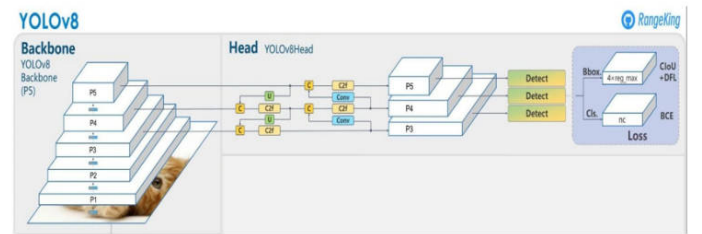


Fig. 3: YOLOv8 architecture [26].

F. Simulation environment and parameter setting

Training the model requires considerable computing power. For this reason, we used Google's Colab environment, which offers access to powerful GPUs for intensive parallel calculations. The hardware configuration was as follows:

- **Processor:** Intel Xeon at 2.20 GHz
- **RAM memory:** 13 GB RAM
- **Graphics card:** NVIDIA T4 Tensor Core GPUs

We used a POCO X4 pro smartphone equipped with a 108-megapixel wide-angle camera to capture videos and images. To carry out this work, several libraries were used, namely: PyTorch, Roboflow, Ultralytics, and Supervision. Specific details regarding the hyperparameter setting used during the training process are outlined in Table I. The choice of these hyperparameters was guided by empirical validation and prior works in object detection.

TABLE I: Hyperparameters used during the training process

Parameters	Value
Epochs	100
Learning rate	0.001
Patience	50 epochs
IoU threshold	0.7
Image size	640
Batch size	16

G. Performance metrics

Performance evaluation of the YOLOv8-based object detection model is carried out using the evaluation metrics described below to see how well it is able to detect and locate different fruits. To understand these metrics, it is important to define True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN). TP refers to the number of fruits correctly detected by the model. TN represents the number of fruits that were correctly missed. FP indicates the number of fruits wrongly detected, while FN represents the number of fruits wrongly missed. Intersection-over-Union measures the overlap between the predicted bounding boxes and the ground truth bounding boxes.

$$IoU = \frac{Area of Overlap}{Area of Union} \quad (1)$$

Precision: Precision measures the proportion of correct detections among all the detections made by the model that are presumed to be correct. It is a fundamental metric for assessing the overall accuracy of predictions.

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

Recall: Recall evaluates the proportion of real objects correctly detected by the model. It gives us an idea of the model's ability to avoid missing objects.

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

Mean Average Precision: The mAP provides an overall assessment of the detection model's performance. Mean Average Precision at IoU:0.5 (mAP@0.5) calculates the average precision across different object categories when considering a detection as correct if the IoU value is equal to or greater than 0.5. Mean Average Precision at IoU:0.5-0.95 (mAP@0.5-0.95) is an extended metric that considers a range of IoU values from 0.5 to 0.95. It calculates the average precision across different object categories, taking into account detections with varying levels of overlap with the ground truth bounding boxes.

$$mAP = \frac{1}{N} \sum_{i=1}^N AP_i \quad (4)$$

where AP_i is the AP value with class index value i , and N denotes the number of classes.

Mean absolute percentage error (MAPE) was also used to measure the detection and counting accuracy of the model. It is defined as:

$$MAPE = \frac{1}{n} \sum_{t=1}^N \frac{|A_t - F_t|}{|A_t|} \quad (5)$$

where A_t is the actual value and F_t is the forecast value. Their difference is divided by the actual value. The absolute value of this ratio is summed for every forecasted point in time and divided by the number of data points.

III. RESULTS AND DISCUSSION

A. Model training and validation

The results of training the YOLOv8 model are shown in Fig. 4 and 5, and Table II.

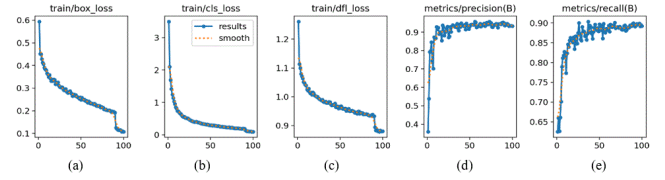


Fig. 4: Model training results.

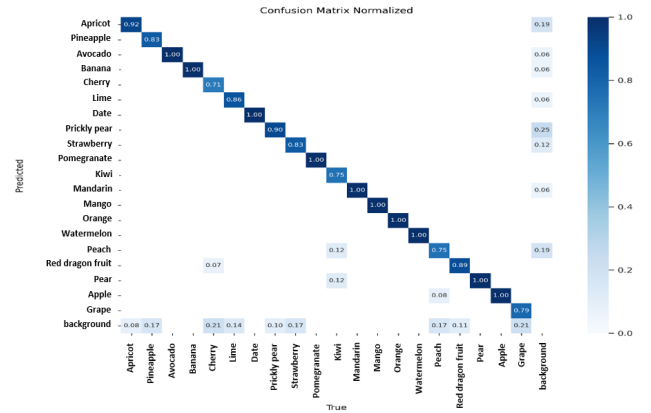


Fig. 5: Confusion matrix of the trained model.

The metric curves illustrated in Fig. 4 represent the progression of the model learning process. Fig. 4 (a), Fig. 4 (b) and Fig. 4 (c) describe the different loss curves of the YOLOv8 model throughout the learning process. The train/dfl_loss curve shows a tendency to reduce losses as the 100 training epochs for fruit classification progress.

The train/class_loss curve decreases over the epochs, stopping at a value of 0.14, indicating that the model accurately predicts fruit classes. The train/box_loss curve also decreases over epochs and stops at a value of 0.13, indicating that the model accurately predicts fruit locations. Fig. 4 (d) and Fig. 4 (e) show the curves for recall and precision of predictions. We note that both curves increase and converge after 50 epochs, reflecting the model's ability to avoid false predictions and detect the majority of fruits.

TABLE II: mAP@0.5 and mAP@0.5-0.95 performance parameter results

Class	mAP@0.5	mAP@0.5-0.95
All	0.942	0.872
Apricot	0.821	0.741
Pineapple	0.972	0.928
Avocado	0.973	0.848
Banana	0.978	0.629
Cherry	0.900	0.848
Lime	0.905	0.842
Date	0.995	0.952
Prickly pear	0.920	0.754
Strawberry	0.820	0.704
Pomegranate	0.995	0.995
Kiwi	0.995	0.939
Mandarin	0.995	0.995
Mango	0.995	0.995
Orange	0.995	0.931
Watermelon	0.995	0.851
Peach	0.857	0.793
Red dragon fruit	0.917	0.878
Pear	0.993	0.997
Apple	0.995	0.995
Grape	0.867	0.854

TABLE III: Model validation results for all classes in the dataset

Class	P	R	mAP@0.5	mAP@0.5-0.95
All	0.915	0.876	0.942	0.877
Apricot	0.703	0.417	0.821	0.741
Pineapple	0.971	0.833	0.972	0.928
Avocado	0.895	0.949	0.973	0.847
Banana	0.875	1.000	0.978	0.630
Cherry	1.000	0.663	0.895	0.844
Lime	0.922	0.849	0.905	0.842
Date	0.979	1.000	0.995	0.952
Prickly pear	0.689	0.900	0.920	0.751
Strawberry	0.816	0.749	0.820	0.704
Pomegranate	0.959	1.000	0.995	0.995
Kiwi	0.974	0.750	0.995	0.939
Mandarin	0.883	1.000	0.995	0.995
Mango	0.989	1.000	0.995	0.995
Orange	0.966	1.000	0.995	0.931
Watermelon	0.950	1.000	0.995	0.923
Peach	0.814	0.750	0.858	0.810
Red dragon fruit	0.981	0.889	0.917	0.881
Pear	0.985	0.960	0.993	0.997
Apple	0.946	1.000	0.995	0.995
Grape	1.000	0.801	0.867	0.854

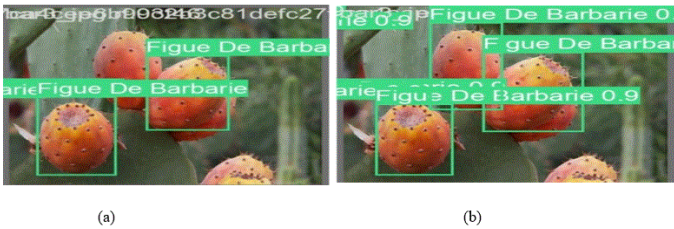


Fig. 7: Model test results. (a) represents the actual bounding boxes and (b) represents the predicted bounding boxes.

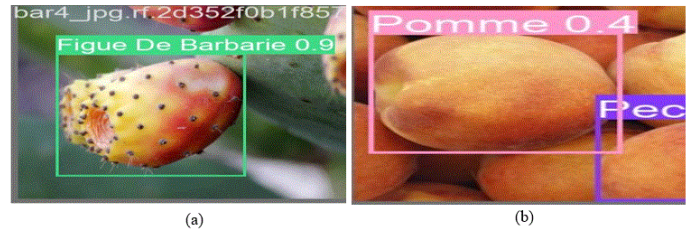


Fig. 8: Errors in (a) Model detection and (b) Classification.

achieved in the recognition of different fruit varieties. For practical application in agriculture, it is essential to accurately

count the number of fruits in an orchard or field. To build a system capable of detecting and counting fruits, we relied on the previously built YOLOv8-based fruit detection and

classification model and the ByteTrack algorithm. This system uses the model we developed during training, capitalizing on its ability to identify and locate fruits in images.

To count fruit, we used the ByteTrack algorithm and a counting tool provided by the Supervision library. ByteTrack enabled us to track each fruit across several frames of a video, guaranteeing accurate counting. It is used to track fruits across different frames of a video. Tracking fruits is essential for accurate fruit counting. This tool associates each fruit detected in one frame with its position in the next frame, enabling fruit movement to be tracked over time. This ensures that each fruit is counted only once, even if it moves from one image to the next.

The supervision library created by the Roboflow team provides a "line counter" counting tool that we used to count the fruits. This tool is responsible for counting how many individual detections have passed a virtual reference line on the video. The line consists of two counters: In, which counts the detected objects that cross the line on the way in, and Out, which counts the detected objects that cross the line on the way out. The in and out line counters analyze information on object detections made and adjust the count accordingly. In this work, if a fruit is detected crossing the line, one of the counters increments the total number of fruits counted according to their direction of travel.

After developing the fruit detection and counting system, it was essential to evaluate its performance on real-world videos. Fig. 9 shows a test video with a simple background, filmed by a smartphone, containing four types of fruits. The counting results are shown in Fig. 10.



Fig. 9: Fruit video without the counting system.



Fig. 10: Fruit video with the counting system.

Fig. 10 shows that the system is able to track four fruits (Apple, Pomegranate, Lime and Banana) and count them with precision, thanks to the counting line. The out counter is

incremented until it reaches the number 4, which corresponds to the actual number of fruits present in the video. Meanwhile, the in counter remains at a zero value, indicating that it counts in the opposite direction of the filmed path. However, some challenges remain, particularly the difficulty of detecting certain fruits due to their hidden position behind the leaves, occlusion, or the position of the fruits too close together (Fig. 11). Addressing these challenges could be a focus for our future work.

Table IV presents a comprehensive summary of the outcomes obtained from assessing the performance of the detection and counting model across five different videos. The results show that the model achieves an accuracy of 75.33% alongside an MAPE value of 24.66%. Compared to previous studies, our system achieves competitive performance. While prior works [22]- [25] mainly focus on one type of fruit (e.g., tomatoes, pears, apples), our approach addresses a more challenging scenario with 20 fruit classes under diverse backgrounds.

TABLE IV: Detection and counting results

Video number	Manual count	Predicted count	Difference
1	10	6	4
2	3	1	2
3	4	4	0
4	5	5	0
5	6	5	1
Total	28	21	7
MAPE	24.66%		
Accuracy	75.33%		

IV. CONCLUSION

In this paper, we presented a real-time system that relies on computer vision and artificial intelligence to detect, classify, and count fruit on a farm automatically and accurately, representing a crucial step in the development of automated information harvesting. As a result, farmers can obtain precise information on the quantity and variety of fruits present in their crops in real time, helping them to make informed decisions about maximizing farm profitability. The system also aims to improve farmers' working time and reduce the workforce required. The system proposed in this paper is based on YOLOv8 and ByteTrack for real-time object detection, tracking, and counting. To develop the system, a dataset containing 20 fruit classes was built with different backgrounds and lighting conditions. Our fruit detection and counting system offers promising results for optimizing agricultural practices by enabling accurate crop assessment. However, it should be noted that improvements are still envisaged, in particular to resolve the problems of erroneous detection and occlusion. For the future, we will pursue research in this field by exploring advanced image processing techniques, expanding the training dataset, and evaluating cross-domain generalization (orchard, market, and field conditions) for better robustness. We will also extend the system to handle more challenging scenarios such as partially occluded, clustered, or damaged

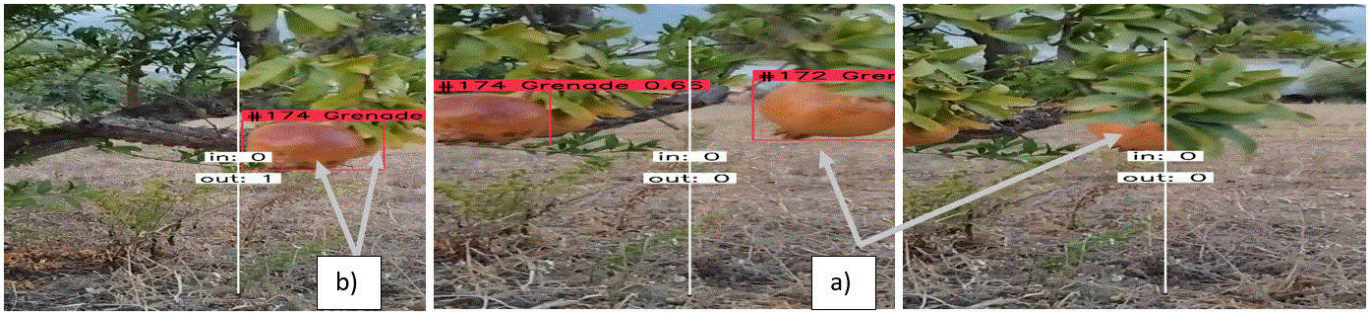


Fig. 11: Fruit video with the counting system.

fruits, thereby enabling recognition of fruits in different states and positions. Moreover, since the counting accuracy remains modest, advanced multi-object tracking and re-identification strategies will be investigated to further enhance counting reliability. Furthermore, a comprehensive comparative evaluation with alternative pipelines such as YOLOv7 + DeepSORT and regression-based counting methods will be conducted. In addition, the integration of this system with drones for crop monitoring and management could be an exciting prospect to explore. Ultimately, this work paves the way for potentially revolutionary applications of deep learning in agriculture, contributing to the realization of tomorrow's intelligent and sustainable agriculture.

REFERENCES

- [1] United Nations, "World Population Projected to Reach 9.8 Billion in 2050, and 11.12 billion in 2100," 2017. [Online]. Available: <https://www.un.org/development/desa/en/news/population/world-population-prospects-2017.html>. [Accessed: Jan. 2024].
- [2] H. Ait Issad, R. Aoudjit, and J. J. Rodrigues, "A comprehensive review of Data Mining techniques in smart agriculture," *Engineering in Agriculture, Environment and Food*, vol. 12, no. 4, pp. 511-525, 2019.
- [3] P. Pawara, A. Boshchenko, L. Schomaker, and M. A. Wiering, "Deep Learning with Data Augmentation for Fruit Counting," in *Proc. of the 19th International Conference on Artificial Intelligence and Soft Computing*, 2020, pp. 203-214.
- [4] C. Wang, Y. Tang, X. Zou, W. SiTu, and W. Feng, "A robust fruit image segmentation algorithm against varying illumination for vision system of fruit harvesting robot," *Optik*, vol. 131, pp. 626631, 2017.
- [5] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 39, pp. 11371149, 2017.
- [6] J. Wu, Z. Kuang, L. Wang, W. Zhang, and G. Wu, "Context-aware RCNN: A baseline for action detection in videos," in *Proc. of the 16th European Conference on Computer Vision*, 2020, pp. 440456.
- [7] R. Girshick, "Fast R-CNN," in *Proc. of the IEEE International Conference on Computer Vision*, 2015, pp. 14401448.
- [8] K. He, G. Gkioxari, P. Dollr, and R. Girshick, "Mask R-CNN," in *Proc. of the IEEE International Conference on Computer Vision*, 2017, pp. 2961-2969.
- [9] W. Liu, D. Anguelov, D. Erhan, C. Szegedy, S. Reed, C. Y. Fu, and A. C. Berg, "SSD: Single Shot MultiBox Detector," *arXiv:1512.02325*, 2016.
- [10] P. Jiang, D. Ergu, F. Liu, Y. Cai, and B. Ma, "A Review of Yolo algorithm developments," *Procedia Computer Science*, vol. 199, pp. 10661073, 2022.
- [11] M. Afonso, H. Fonteijn, F. S. Fiorentin, D. Lensink, M. Mooij, N. Faber, G. Polder, and R. Wehrens, "Tomato Fruit Detection and Counting in Greenhouses Using Deep Learning," *Frontiers in Plant Science*, vol. 11, p. 571299, 2020.
- [12] F. Gao, L. Fu, X. Zhang, Y. Majeed, R. Li, M. Karkee, and Q. Zhang, "Multi-class fruit-on-plant detection for apple in SNAP system using Faster R-CNN," *Computers and Electronics in Agriculture*, vol. 176, p. 105634, 2020.
- [13] G. Liu, J. C. Nouaze, P. L. T. Mbouembe, and J. H. Kim, "YOLO-Tomato: A Robust Algorithm for Tomato Detection Based on YOLOv3," *Sensors*, vol. 20, no. 7, p. 2145, 2020.
- [14] G. Yang, J. Wang, Z. Nie, H. Yang, and S. Yu, "A Lightweight YOLOv8 Tomato Detection Algorithm Combining Feature Enhancement and Attention," *Agronomy*, vol. 13, no. 7, p. 1824, 2023. DOI: 10.3390/agronomy13071824.
- [15] C. Song, K. Wang, C. Wang, Y. Tian, X. Wei, C. Li, Q. An, and J. Song, "TDPPL-Net: A lightweight real-time tomato detection and picking point localization model for harvesting robots," *IEEE Access*, vol. 11, pp. 37650-37664, 2023.
- [16] A. Bewley, Z. Ge, L. Ott, F. Ramos, and B. Upcroft, "Simple Online and Realtime Tracking," *arXiv:1602.00763*, 2016.
- [17] N. Wojke, A. Bewley, and D. Paulus, "Simple Online and Realtime Tracking with a Deep Association Metric," *arXiv:1703.07402*, 2017.
- [18] Y. Zhang, C. Wang, X. Wang, W. Zeng, and W. Liu, "FairMOT: On the Fairness of Detection and Re-Identification in Multiple Object Tracking," *arXiv:2004.01888*, 2020.
- [19] Y. Zhang, P. Sun, Y. Jiang, D. Yu, F. Weng, Z. Yuan, P. Luo, W. Liu, and X. Wang, "Bytetrack: Multi-object tracking by associating every detection box," *arXiv:2110.06864*, 2022.
- [20] A. Cioppa, S. Giancola, A. Deliege, L. Kang, X. Zhou, Z. Cheng, B. Ghanem, and D. Droogenbroeck, "SoccerNet-Tracking: Multiple Object Tracking Dataset and Benchmark in Soccer Videos," in *Proc. of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 3491-3502.
- [21] M. N. J. Attzs, T. Mahendrakar, M. J. Meni, R. T. White, and I. Silver, "Comparison of Tracking-By-Detection Algorithms for Real-Time Satellite Component Tracking," 2023.
- [22] Y. Egi, M. Hajyzadeh, and E. Eyceyurt, "Drone-Computer Communication Based Tomato Generative Organ Counting Model Using YOLO V5 and Deep-Sort," *Agriculture*, vol. 12, p. 1290, 2022.
- [23] W. Zhang, J. Wang, Y. Liu, K. Chen, H. Li, Y. Duan, W. Wu, Y. Shi, and W. Guo, "Deep-learning-based in-field citrus fruit detection and tracking," *Horticulture research*, vol. 9, 2022.
- [24] A. I. B. Parico and T. Ahamed, "Real Time Pear Fruit Detection and Counting Using YOLOv4 Models and Deep SORT," *Sensors*, vol. 21, no. 14, p. 4803, 2021.
- [25] J. Hu, C. Fan, Z. Wang, J. Ruan, and S. Wu, "Fruit Detection and Counting in Apple Orchards Based on Improved YOLOv7 and Multi-Object Tracking Methods," *Sensors*, vol. 23, no. 13, p. 5903, 2023.
- [26] N. Buhl, "YOLO models for Object Detection Explained," [Online]. Available: <https://encord.com/blog/yolo-object-detection-guide/>. [Accessed: Jan. 2024].
- [27] M. Hussain, "YOLO-v1 to YOLO-v8," *Machines*, vol. 11, p. 677, 2023.
- [28] G. Wang, Y. Chen, P. An, H. Hong, J. Hu, and T. Huang, "UAV-YOLOv8," *Sensors*, vol. 23, p. 7190, 2023.